

1. Train Test Split

Train Test Split is a technique used to divide a dataset into two parts:

1. **Training Data**
2. **Testing Data**

Training Data

- Used to **train the machine learning model**.
- Usually **70% – 80%** of the dataset.

Testing Data

- Used to **evaluate the model performance**.
- Usually **20% – 30%** of the dataset.

Example

If a dataset has **1000 rows**:

- **800 rows** → **Training**
- **200 rows** → **Testing**

Purpose

- Prevent **overfitting**
- Check how the model performs on **unseen data**

Formula

Dataset = Training Data + Testing Data

In Python, it is implemented using **Scikit-learn**.

2. Correlation

Correlation measures the **relationship between two variables**.

It shows how strongly two features are related.

Correlation Values

Value	Meaning
+1	Perfect positive correlation
0	No correlation
-1	Perfect negative correlation

Types of Correlation

1. Positive Correlation

- When one variable increases, the other also increases.

Example:

- Study hours $\uparrow \rightarrow$ Marks $\uparrow$

2. Negative Correlation

- When one variable increases, the other decreases.

Example:

- Price $\uparrow \rightarrow$ Demand $\downarrow$

3. Zero Correlation

- No relationship between variables.

Correlation Matrix

A **correlation matrix** shows correlation values between all features in a dataset.

Visualization can be done using ****Seaborn heatmap**.

Importance

- Helps in **feature selection**
- Removes **highly correlated features**
- Improves **model performance**

3. Feature Selection

Feature Selection is the process of selecting the **most important features** from a dataset for training a machine learning model.

Why Feature Selection is Important

- Reduces **overfitting**
- Improves **model accuracy**
- Reduces **training time**
- Makes the model **simpler and efficient**

Types of Feature Selection Methods

1. Filter Method

- Uses statistical techniques to select features.
- Example:
 - Correlation
 - Chi-Square test
 - Variance Threshold

2. Wrapper Method

- Uses machine learning algorithms to select features.

Example:

- Forward Selection
- Backward Elimination
- Recursive Feature Elimination (RFE)

3. Embedded Method

- Feature selection happens during model training.

Example:

- Lasso Regression
- Random Forest Feature Importance

Example

In a **house price prediction dataset**, features may include:

- Area
- Number of rooms
- Location
- Age of house

Feature selection helps choose the **most relevant features** for prediction.